

```
#!/bin/perl
```

```
my ($subject, $mname, $from, $to, $attachment, $fh, $content);
$subject = "Test Mail";
$from = "osfyuser";
$to = "osfyedit";
$attachment="/home/osfyuser/test.txt";
```

```
$fh="mailbody.txt";
open(FILE, ">", "$fh") or die "Cannot open $fh: $!";
print FILE "This is a text for all mailers\welcome to the
world of mailing... Buhahaha!!";
close(FILE);
```

```
open(MAIL, "|/usr/sbin/sendmail -t");
print MAIL "FROM: $from\n";
print MAIL "TO: $to\n";
print MAIL "Subject: $subject\n";
print MAIL "Content-Type: multipart/mixed;
boundary=frontier\n";
print MAIL "--frontier\n";
print MAIL "Content-Type: text/plain; charset=us-ascii\n\n";
open(FILE, "<", "$fh") or die "Cannot open $fh: $!";
print MAIL <FILE>;
close(FILE);
print MAIL "\n\n";
print MAIL "--frontier\n";
chomp(my $basename='basename $attachment');
print MAIL "Content-Disposition: attachment;
filename=$basename\n";
print MAIL "Content-Type: text/plain; name=$attachment\n\n";
open(FILE, "<", "$attachment") or die "Cannot open
$attachment: $!";
print MAIL <FILE>;
print MAIL "\n\n";
close(FILE);
close(MAIL);
```

For sending .pdf, .zip or .doc or .xlsx files, we need to use:

- i) uuencode or base64 for Shell
- ii) base64 and the IO::File module for Perl

The following are only small snippets we can adjust/add to the code provided earlier.

For Shell, the snippet is:

```
echo "Content-Transfer-Encoding: uuencode"
#echo "Content-Transfer-Encoding: base64"
echo -e "Content-Disposition: attachment; filename='basename
$attachment'"
echo -e "Content-Type: application/octet-stream;
name=$attachment\n";
uuencode $attachment 'basename $attachment'
#base64 $attachment
```



Figure 5: Mail with attachment seen with browser mail client

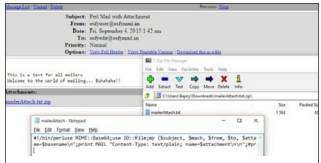


Figure 6: Attachment opened from the browser using Zip utility

For Perl, the snippet is:

```
#print MAIL "Content-Transfer-Encoding: base64\n";
#print MAIL "Content-Transfer-Encoding: uuencode\n";
print MAIL "Content-Disposition: attachment;
filename=$basename\n";
print MAIL "Content-Type: application/octet-stream;
name=$attachment\n\n";
# print MAIL read_file($attachment); # Can be also used for
text/plain attachments
print MAIL encode_base64( read_file($attachment) ); # CTE must
be 64; this must be accompanied by the MIME::Base64 module
which provides this function for encoding. It comes default
on all Perl installs.
#print MAIL 'base64 $attachment'; # CTE must be 64; this
procedure is incase you do not want to use the above module.
Using the Shell command ticks.
#print MAIL 'uuencode $attachment $basename'; # CTE must
be uuencode; this also uses the Shell command ticks to use
uuencode instead of base64
```

The following function is required for Perl to read the file properly, only if we use the `encode_base64` function from the `MIME::Base64` module. We could also use this function for plain text as in a commented portion in the code earlier.

```
sub read_file
{
    my $filename = shift @_;
    my $attach = new IO::File;
    $attach->open("< $filename")
        or die "Error opening $filename for reading - $!\n";
```